ECEN 607: Advanced Analog Circuit Tech

Lab 1: MOS Device Characterization

Lab Report

Name: FirstName LastName
UIN: ******Last Three Digits

Objectives:

To review concepts learned in background classes such as characterization, small-signal and CMOS circuit design.

5 V Transistor Characterization

High voltage CMOS design is done in many applications even today. In this course, you will be designing circuits operating both at 1.8 V supply and 5.5 V supply. Using the technology process IBM180 and the following procedure, characterize the four terminal, **5.5 V** nominal, MOS transistors **nfetmx** and **pfetmx**.

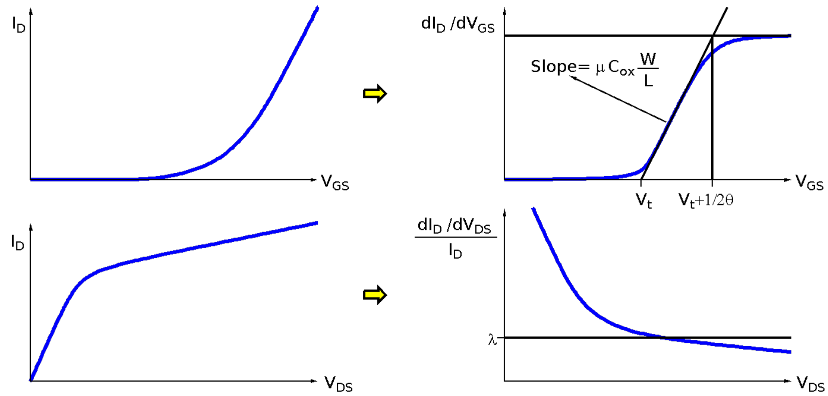
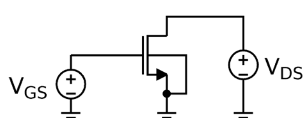


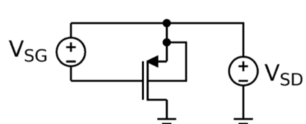
Figure 1-11: Curves used for CMOS characterization

To obtain g_m, derivative of I_D needs to be plotted as a function of V_{GS}. The resulting plot can be used to extract V_t, μC_{ox} and θ, as shown above. Channel length modulation parameter lead to the finite value of the output conductance, and can be extracted by taking the derivative of I_D as a function of V_{DS}.

1. Construct the circuit below using **nfetmx** transistor with W/L=1μ/0.7μ.



   a. Fix V_{DS}=2V. Sweep V_{GS} from 0 to 5V, plot I_D and gm (dI_D/dV_{GS}) vs. V_{GS}, and extract V_{tn}, $\mu_n C_{ox}$ and θ.
   b. Set $V_{GS}=V_{tn}$+100mV. Sweep V_{DS} from 0 to 5V, plot and gds (dI_D/dV_{DS}) vs. V_{DS}. Plot the output resistance rds, and its first and second derivative as function of V_{DS}.
   c. Find the intrinsic gain (A_i) and unity-gain frequency (f_t) using AC simulations when V_{DS}=2V and $V_{GS}=V_{tn}$+100mV.
2. Repeat (1) for the **nfetmx** transistor with W/L=5μ/0.7μ.
3. Repeat (1) for the **nfetmx** transistor with W/L=5μ/1.4μ.
4. Construct the circuit below using **pfetmx** transistor with W/L=1μ/0.5μ.



   a. V_{SD}=2V. Sweep V_{SG} from 0 to 5V, plot I_D and gm=dI_D/dV_{SG} vs. V_{SG}, and extract V_{tp}, $\mu_p C_{ox}$ and θ.
   b. $V_{SG}=|V_{tp}|$+100mV. Sweep V_{SD} from 0 to 5V, plot and gds (dI_D/dV_{SD}) vs. V_{DS}. Plot the output resistance rds, and its first and second derivative as function of V_{SD}.

c. Find the intrinsic gain (A_i) and unity-gain frequency (f_t) using AC simulations when V_{SD}=2V and V_{SG}=$|V_{tp}|$+100mV.

5. Repeat (4) for the **pfetmx** transistor with W/L=5µ/0.5µ.
6. Repeat (4) for the **pfetmx** transistor with W/L=5µ/1µ.

In the lab report, include your schematics, plots, annotations, and calculations. Please provide the characterization results in a plots and comment on the results.